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Solvent effects on the wood delignification with sustainable solvents

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ABSTRACT

Solutions of a pure organic solvent acidified with 1% sulfuric acid, and some of their aqueous mixtures were used for lignin extraction in the *Pinus radiata* sawdust delignification. Organic acid solvents including acetic, lactic and citric acids as well as non-acidic compounds such as γ-valerolactone, 2-methyltetrahydrofuran, glycerol and ethylene glycol were studied. Crude lignin extractions yields (%) ranging between ca. 5–50% were obtained, from which ethylene glycol (33%), γ-valerolactone (48%) and propylene carbonate (52%) showed the greatest effectiveness. The effect of added water on the lignin extraction was investigated in mixtures of an organic solvent with a variable water content (75%*w*/w, 50% w/w, 25%w/w and 10% *w*/w) where it was observed that the yield of extraction decreased with the increased water content. Moreover, the purity of extracted lignins were analyzed by spectroscopic methods (UV and IR). Kamlet-Taft solvent polarity parameters, were determined with the solvatochromic probes 4-nitroaniline, *N,N*-diethyl-4-nitroaniline, Nile Red and 6-propionyl-2-*N,N*-dimethylaminonaphthalene (PRODAN), and then correlated to lignin extraction yields to explain the influence of the solute-solvent interactions on biomass delignification. A reasonable correlation was found between the medium polarizability-dipolarity π* and the effectiveness of the solvent mixture on the extraction of lignin wood.

1. Introduction

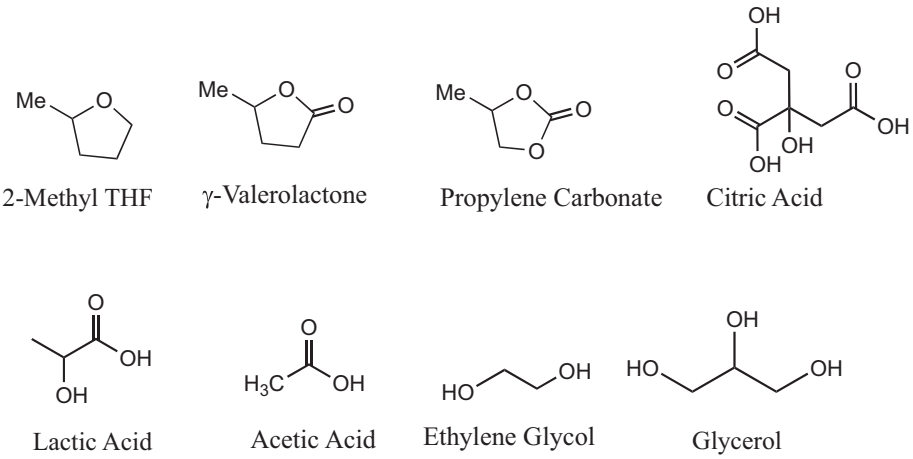
Lignocellulosic biomass is widely recognized as an important renewable resource from which it is possible to produce bioenergy, novel materials, and platform for value-added chemical products [1]. The main components of lignocellulosic biomass are ascribed to lignin (ca. 20%), cellulose (ca. 40%), and hemicellulose (ca. 25%) which are not uniformly distributed within the cell walls. The structure and the amount of these cell-wall components vary according to species, types (soft-wood, hardwood or grasses), tissues and maturity of the plant biomass, factors that affect their amenability to deconstruction [2,3].

Commonly, the separation process of biomass into their constituents (i.e., cellulose, hemicellulose and lignin), is carried out by biological, physical or chemical methods. Regardless of the method, a pre-treatment is a fundamental step which breaks down the network formed by hemicellulose and lignin and exposes the plant fiber to an efficient hydrolysis that leads to products with an added value [4]. Isolated lignin possesses structural characteristics that depend on the delignification process. In general, as result of the chemical cleavage of abundant aryl ether bonds in lignin structure, a high content of phenolic hydroxyl groups which are responsible for its antioxidant activity [5]. Lignin and lignin-related polymers have been reported as additives in biodegradable packaging materials, [6] as food additives, [7] and in cosmetics [8]. Conventional pre-treatment methods for biomass deconstruction include Kraft, Sulphite and Organosolv pulping, which are performed at high temperatures (120–180 °C) by using strong alkali, sulfur chemicals and enormous amounts of solvents, that have a negative environmental impact. In consequence, the fractionation of the main components of lignocellulosic biomass particularly the extraction and isolation of lignin by more sustainable methodologies constitute a present challenge in biomass valorization [9].

Sustainable chemistry focuses on the design of chemical processes that cause minimal environmental impact, seeking to reduce and prevent pollution at its source [10,11]. It is well accepted that a major problem in the fine and chemical industries arises from the use of organic solvents as a major source of waste which is associated with

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Scheme 1. Solvents used in this work.

serious health hazards and/or environmental problems. Nevertheless, solvent plays a fundamental role in chemical processes. Hence, the need for replacement of conventional organic solvents (COS) by suitable alternative solvents is widely recognized and has become one of the major focus of current research in industry and academy [12,13].

Several approaches have been used for developing sustainable biomass fractionation processes. The use of novel and greener solvents in the biomass fractionation process could increase lignin extraction and enhance its physicochemical characteristics (less lignin degradation) for added value applications, instead of its current combustion to produce energy. In this context, neoteric solvents such as ionic liquids (ILs), deep eutectic solvents (DES) and their aqueous mixtures have emerged as alternatives to conventional solvents employed in lignocellulosic biomass processes [14]. The so-called bio-based solvents (BBSs), [15] obtained from renewable biomass, as well as from animal or mineral raw materials, have been used as solvents in some organic processes and have shown to be effective without compromising the safety and the environmental acceptability of the process, being therefore compatible with sustainable development [16–18]. Among them, glycerol, 2-methyltetrahydrofuran (2-MeTHF), γ-valerolactone (GVL) and lactic acid derivatives are the most used (Scheme 1). Their main advantages over other solvents include their low toxicity and biocompatibility, besides the fact that they are obtained without consumption of fossil resources. They have been used in the pretreatment of biomass with excellent results.

For example, Zhang et al used glycerol carbonate and ethyl carbonate as solvents for the pre-treatment of sugarcane bagasse fractionation, showing that acidified glycerol carbonate was effective to breakdown lignocellulosic biomass and further allowing lignin extraction. The researchers attributed these results to the presence of one hydroxyl group in the glycerol structure, resulting in more significant biomass delignification [19]. In another study, treatment of sugarcane bagasse by glycerol removed >65 wt% lignin and >80 wt% hemicellulose while keeping cellulose (>95 wt%) almost intact, the latter being efficiently hydrolyzed (>90 wt% conversion) after 24 h [20]. Calcio et al. [21] used a GVL/water mixture for wheat straw delignification obtaining 27% of lignin extraction yield under ultrasound irradiation. Recently, Pang et al. explored the delignification process with aqueous mixtures of *p*-toluenesulfonic acid (p-TSA) with different biomass samples, observing higher efficiencies when using hardwood biomass [22].

Empirical solvent parameters provide an adequate framework to interpret solute–solvent interactions in chemical processes. Solvatochromic probes have been used to characterize medium polarities and are widely used in correlations with experimental results. In particular, Kamlet-Taft (K-T) parameters have been related to the solvent performance in biomass delignification and extraction of target molecules [23,24]. Oh et al. [25] performed pine wood delignification with different types of cholinium chloride-based deep-eutectic-solvents (DES) and found that the hydrogen-bond acidity (K-T parameter α) of the solvent showed a linear relationship with the lignin extraction efficiency. On the other hand, Liu et al. [26] and Hong et al. [27] concluded that the lignin-dissolving ability correlated well with the difference between the values of K-T α and β (α-β) of DESs. Other studies have also shown a close relation between lignin extraction/solubility and π* values [28,29]. Nevertheless, in spite of these reports, there is still a lack of systematic work relating solvent parameters and lignin extraction efficiency.

In the present work, several bio-based solvents (see scheme 1) and their acidic aqueous mixtures were employed in the extraction of lignin from *Pinus radiata* sawdust. These acidic mixtures play a crucial role in the delignification process: they promote the breakdown of the polymeric network into more soluble fragments leaving exposed the plant fiber to then efficiently remove the lignin from the insoluble residues. Thus, their effectiveness on the lignin extraction reflects their ability as promoting agents for breakdown and as solubilizing media for the resulting fragments. Because biomass delignification and lignin extraction are concomitant processes, in the present work we have employed both terms – extraction and reaction – with little discrimination, when describing the medium effectiveness. Nevertheless, since the wood delignification involves the two processes, we have distinguished in the experimental section, the pretreatment and biomass delignification, where chemical reactions happens with the cleavage of ether bonds of a polymeric material to produce smaller fragments, from the lignin extraction, where these fragments are extracted by the solvent.

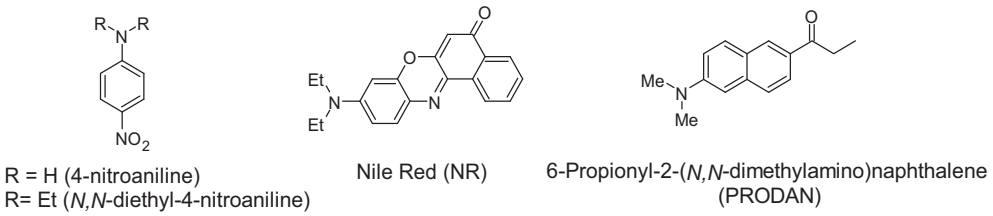
The overall solvent effectiveness was evaluated and compared by the degree of recovery of extracted lignin after the pretreatment process. A simple spectrophotometric method was used to estimate the purity of the extracted lignin. In addition, solvent parameters were determined, with the aid of solvatochromic probes, to explain the solvent effectiveness and the water content effect on the lignin extraction.

Scheme 1 lists all solvents employed in the present work.

2. Experimental

2.1. Materials

2-Methyltetrahydrofuran, lactic acid, γ-valerolactone, and glycerol were purchased from Merck and Sigma-Aldrich. Citric acid and ethylene glycol were purchased from Chemix. 4-nitroaniline, and *N,N*-diethyl-4-nitroaniline were purchased from Sigma-Aldrich. Nile Red and PRODAN were purchased from Santa Cruz Biotechnology. A MilliQ Millipore system was used for the purification of water (resistivity <18 MΩ). *Pinus*



Scheme 2. Solvatochromic probes employed for solvent polarity measurements

radiata sawdust was obtained from a local sawmill, was air-dried, ground and sieved (0.18–0.85 mm, −20 to +80 of US mesh scale) before use. The biomass was then stored in plastic bags at room temperature.

2.2. Biomass delignification

0.5 g of dry biomass and 10.0 g of solvent were mixed into a 50 mL Teflon-lined stainless-steel autoclave (reactor), the reactor was sealed (no additional pressure) and placed in an oil-bath, pre-heated and once the temperature reached 110 °C, the reactor was left for 5 h under continuous magnetic stirring. Afterwards, the reactor was cooled down to room temperature.

2.3. Lignin extraction

After biomass delignification process, the resulting solid and liquid phases were separated by filtration under vacuum through 0.45 μm nylon membrane filters (Merck Millipore, USA). The recovered solid biomass was washed with 100 mL of ultrapure H_2O (in 10 mL portions) to ensure removal of the solvent, while liquid phase was collected and 30 mL of mQ water were added to induce lignin precipitation. The resulting solution was stored in a freezer for 24 h to increase the efficiency of lignin precipitation. The resultant suspension was filtered under vacuum through 0.45 μm nylon membrane filters to recover lignin as solid material. All filtered solids were placed in an oven at 60 °C for 24 h and afterwards were stored at room temperature. All experiments were performed at least in duplicate, with average results showing less than 5% variation.

2.4. Solvent parameters

Stock solutions (2 mM) of 4-nitroaniline and N,N-diethyl-4-nitroaniline and 500 mM of Nile Red and PRODAN were prepared by dissolving them in ethanol in precleaned amber vials and stored at 4 ± 1 °C. 50 μL of the probe solution was transferred from the stock solution to a 1-mm path quartz cuvette. The ethanol was evaporated, the dry residue was dissolved in the desired solvent to a final volume of 2.5 mL for fluorescence and 500 μL for absorbance experiments. A Jasco V-630 spectrophotometer was used for the acquisition of the UV–Vis absorbance data. The fluorescence spectra were acquired on a Jasco equipment. All absorbances were measured using quartz cuvettes of optical path 0.2 cm while fluorescence data were measured using 1-cm quartz cuvettes at 25 °C. All spectroscopic measurements were performed in triplicate.

Kamlet-Taft solvent parameters were determined with a set of solvatochromic probes that included 4-nitroaniline and *N*,*N*-diethyl-4-nitroaniline. The use of Reichardt's $E_T(30)$ betaine for the determination of the hydrogen-bond-donor strength of the medium was precluded by the acidic conditions of the solvents. To replace it by other acid-insensitive probes, we resorted to solvatochromic Nile Red and fluorosolvatochromic PRODAN, both of them exhibiting a positive solvatochromism that correlated well with the $E_T(30)$ scale. The structures of all probes employed in the present work are shown in Scheme 2.

2.5. IR measurements

The analyses of the samples were carried out using a Jasco FT-IR system equipped with a single ATR cell (attenuated total reflectance), and a diamond crystal. All data were recorded at room temperature, in the range of 4000–600 cm^{-1} by accumulating 32 scans with a resolution of 4 cm^{-1} and interval of 2 cm^{-1}. At least four spectrum acquisitions were obtained for each sample and temperature. All spectra were baseline corrected for further analysis.

2.6. NMR analysis

2-Dimensional (2D) experiments of heteronuclear ^{1}H—^{13}C measurement were conducted in a Bruker 400 MHz spectrometer (AVANCE II, Bruker, Germany). The samples were prepared by homogeneously mixing the lignin sample (40 mg) with DMSO-d_6 (0.6 mL).

3. Results

3.1. Lignin-extracting efficiency

Pinus radiata sawdust delignification efficiency was compared for different acidic aqueous mixtures of organic solvents which included alcohols (ethylene glycol and glycerol), an ether (2-methyltetrahydrofuran) and a lactone (γ-valerolactone), besides several acids, such as acetic, lactic, and citric acid and finally their corresponding aqueous solutions. In all cases, the acidic medium was ensured by the addition of 1% *w*/w of concentrated sulfuric acid to 10 g of the pure organic solvent, or aqueous mixture.

Two sets of experiments were carried out, the first employing acidified pure organic solvents, and the second, by varying the water content in some of the selected mixtures.

The dried extracts of crude lignin were weighed, to determine the corresponding extraction yields. They were then submitted to NMR, IR and UV analysis, to obtain a clearer picture of their purity.

HSQC spectrum of lignin is usually divided into three regions: the aliphatic, the side-chain and the aromatic region. The aliphatic region provides information about the presence of acetyl groups attached to the lignin polymer at around $\delta C/\delta_H$ 20–21/1.7–1.9 ppm. The side-chain region corresponds to linkages in the lignin structure, and a conspicuous signal at around at $\delta C/\delta_H$ 56.0/3.70 ppm, which is attributed to methoxyl substituents on the aromatic rings. The aromatic region of the HSQC spectra contains correlations relative to guaiacyl, syringyl and *p*-hydroxyphenyl subunits and of ferulic and p-coumaric acid monomers. The HSQC spectrum of the crude lignin obtained from extraction with γ-valerolactone is reproduced in the Supporting Information (Fig. S1) with a list of assigned peaks, shown in Table S1. Comparison with analogous spectra of lignin [30–32] confirmed the chemical nature of the expected product.

IR spectra of the recovered lignin samples confirmed the presence of characteristic aromatic bands in the 1600–1500 cm^{-1} region, in addition to absorptions due to C—H stretching in the 2900–3000 cm^{-1} region. These spectra were very similar to lignin spectra reported in the literature [33,34] (Fig. S2 and S3 in the Supporting Information). However, spectra of lignin obtained with pure γ-valerolactone showed

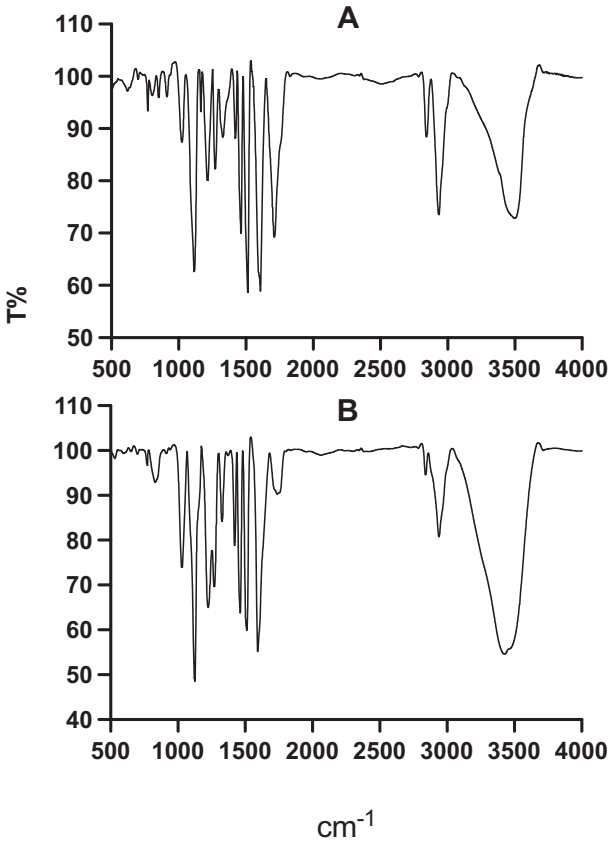


Fig. 1. IR spectra of a sample of crude lignin extracted by γ-valerolactone (1% *w*/w sulfuric acid, 110 °C, 5 h, 500 rpm) (A) before and (B) after repeated washings with water.

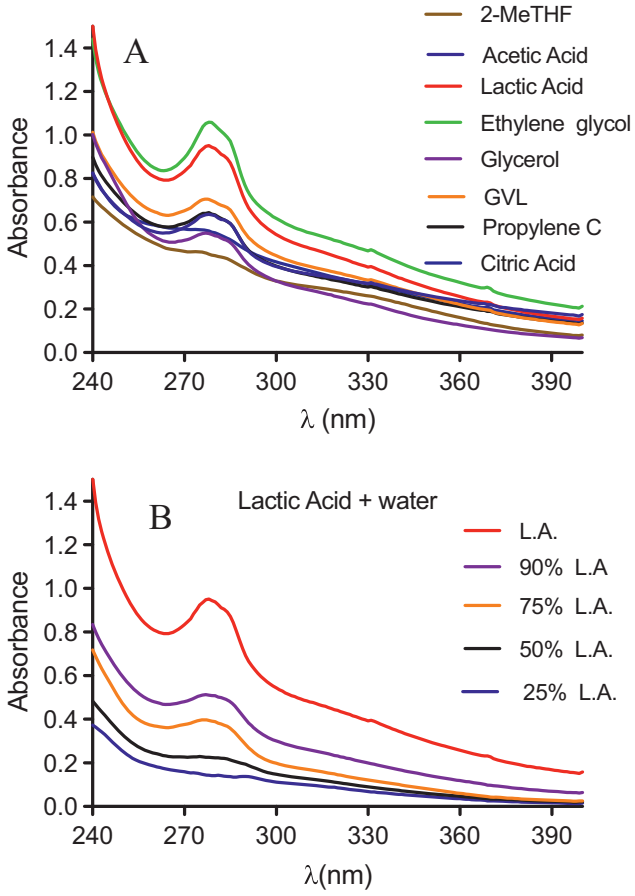


Fig. 2. UV–vis spectra of samples of crude lignin sample in THF (0.1 g L^{-1}) isolated (A) from acidified organic solvents and (B) from Lactic Acid (L.A.) and an increasing content of added water. Experimental conditions of extraction: 1% *w*/w sulfuric acid, 110 °C, 5 h, 500 rpm.

an extra absorption band in the 1650–1700 cm^{-1} region, assigned to a carbonyl stretching. This pointed to the existence of impurities originating from the solvent. Attempts to eliminate these impurities by repeated washings led to their partial elimination, as shown by the decrease of the C═O band intensity (Fig. 1). Nevertheless, it was clear that the lignin extracts consisted of mixtures of pure lignin with variable contents of solvent-derived impurities. Yields based on these "crude" extracts were therefore distinguished from the yields of pure lignin present in these mixtures, which were estimated by their IR and UV–Vis spectra. Yields always refer to the lignin extraction, and are simply a ratio between the solubilized, extracted material and the original biomass. As there is an indefinitely large number of chemical reactions involved in the breakdown of the polymeric network, little can be said about their particular yields.

Taking into account the existence of impurities, more quantitative estimates of the lignin content in the crude extracts were obtained from their UV–Vis spectra. The isolated lignin samples were readily dissolved in tetrahydrofuran (THF) forming clear brown solutions. The UV absorption spectra of the dissolved lignin samples were similar to the one obtained from the dissolved loblolly pine wood or a synthetic guaiacyl lignin made from coniferyl alcohol [35], showing a band at 280 nm, characteristic of aromatic groups. This absorption provided a means to quantify the pure lignin present in crude samples, as the possible impurities were all saturated, aliphatic derivatives. Fig. 2A shows the changes in the absorbance at 280 nm for different samples of crude lignin. UV–vis spectra of crude samples obtained by extraction with mixtures of lactic acid and variable water contents also showed that lignin purity decreased with the increased amount of water in the extracting mixtures (Fig. 2b). These results agreed with the qualitative conclusions from the IR spectra of the same crude samples. In all cases,

Table 1
– Yields of crude and pure lignin obtained by wood delignification with various acidified solvents and their aqueous mixtures.

Extracting solvent	A_{280} [a]	% lignin in crude sample [b]	Crude yield, %	Pure yield, % [c]
Ethylene glycol	1.05	98.4	33	32.5
Lactic acid	0.95	88.3	21.7	19.2
Lactic acid + water (10% *w*/w)	0.5	42.8	50	21.4
Lactic acid + water (25% w/w)	0.38	30.7	57.8	17.7
Lactic acid + water (50% w/w)	0.22	14.5	25.5	3.7
Lactic acid + water (75% w/w)	0.14	6.5	9	0.6
γ-Valerolactone	0.70	63.0	48.1	30.3
γ-Valerolactone + water (10% w/w)	0.5	42.8	45.5	19.5
γ-Valerolactone + water (25% w/w)	0.5	42.8	33.9	14.5
γ-Valerolactone + water (50% w/w)	0.37	29.7	29.1	8.6
Acetic acid	0.63	56	7.8	4.4
Propylenecarbonate	0.63	56	52.5	29.4
Citric acid	0.56	48.9	9	4.4
Glycerol	0.54	46.9	12.5	5.9
2-MeTHF	0.44	36.8	25.3	9.3

[a] Absorbance reading at 280 nm of a THF solution of the crude lignin with a concentration of 0.1 g. L^{-1}.
[b] Calculated as % lignin = (A_{280}–0.076)/0.99.
[c] Calculated as pure yield = % lignin × crude yield.

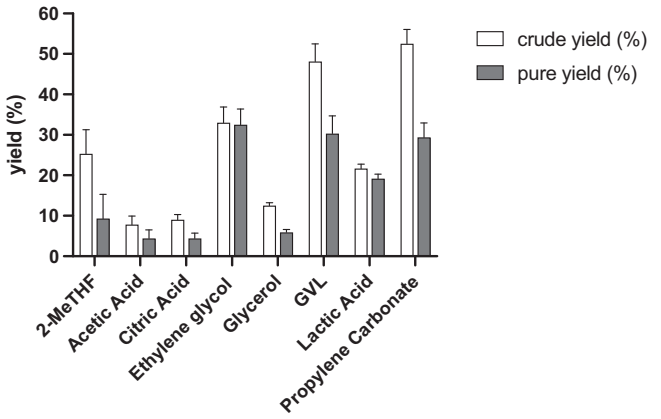


Fig. 3. Crude and pure lignin yields after wood delignification with examined organic solvents. Experimental conditions of extraction: 1% *w*/w sulfuric acid, 110 °C, 5 h, 500 rpm.

the content of pure lignin decreased with the amount of water added to the organic solvent (Fig. S4 in supporting information).

Taking into account the absorbance value of its at 280 nm, the purest sample of extracted lignin was obtained with ethylene glycol. In the absence of a molar absorptivity coefficient for the biopolymeric material, the spectrum in THF of this sample was used as a reference to estimate the amount of pure lignin in the other extracts. A plot regarding the variation of the absorbance A_{280} at 280 nm for different concentrations of the sample in THF yielded a straight line ($r^2 = 0.99$) (Fig. S5 and S6 in the Supporting Information), in agreement with a Beer-Lambert behavior.

In this way, a linear relationship was obtained between the absorbance value A_{280} and the concentration *c* of lignin in all crude 0.1 g dry weight samples, $A_{280} = 0.076 + 9.9\ c$.

This equation allowed the estimation of the percentage of pure lignin in 0.1 g of the crude sample as % lignin = $(A_{280}–0.076)$ / (0.99). The corresponding yields of pure lignin were obtained as the product of the crude yields by % lignin in the samples. Yields for the crude and pure lignin are listed in Table 1 and summarized in Fig. 3 for all extracting solvents and aqueous solvent mixtures.

The yields of extracted lignin in all pure solvents are compared graphically in Fig. 3.

As can be seen in Fig. 3 and from the data of Table 1, biomass delignification with ethylene glycol, γ-valerolactone and propylenecarbonate led to good yields of relatively pure lignin. Reasonable yields were obtained with lactic acid, whereas poor yields were obtained with 2-MeTHF, glycerol, citric and acetic acid.

The addition of water to the organic solvents led in general to poorer yields of more impure products. This can be gathered from the plots of Fig. 4, where the lignin extraction efficiency of aqueous mixtures with lactic acid and γ-valerolactone are compared.

The above results can be analyzed in search of patterns that allow some general questions to be answered. The first of them is related with the comparative ability of different acidic organic solvents to promote delignification. The second addresses the effect of the added water on the delignification performance of organic solvents. Finally, a search of possible correlations between the extraction efficiency and specific medium properties, quantified by Kamlet-Taft parameters, can lead to insights into solute-solvent interactions between lignin and organic solvents and/or water.

As seen in Fig. 4, there were no consistent differences between neutral and acidic organic solvents in promoting delignification. Among the best extracting solvents, neutral propylene carbonate and γ-valerolactone appeared side-by-side with hydroxylic ethylene glycol, followed by lactic acid with an intermediate performance. However, other neutral solvents (glycerol, 2-methyltetrahydrofuran) performed as poorly as more acidic compounds like acetic or citric acid.

The reasons for their different behavior must therefore be sought not in their relative acidities, but also in other factors, as, for example, their chemical stability under the conditions of the hot acid-catalyzed delignification process and the viscosity and polarity of the solvent among others.

The poor performance of citric acid could be attributed to its ready decarboxylation upon heating, a well-known process that takes place even as a solid upon melting. Glycerol is also known to undergo an acid-catalyzed dehydration with the formation of a number of products, [36] being even proposed as a green source of acrolein [37].

Both lactic acid and ethylene glycol polymerize under acidic conditions, in condensation processes that liberate water as a by-product [38]. These processes follow the same mechanistic pathway that should govern the biomass delignification: acid-catalyzed, solvent-assisted nucleophilic substitutions, leading, however, not to the lignin fragmentation, with which it competes, but to the consumption of solvent molecules, the formation of water and of solvent-derived impurities.

In fact, in all the above extractions, the water content in the mixture increased, either due to a reduction of the molecular weight of the organic co-solvent, or because water was formed as a by-product of polymerization reactions. As shown in Fig. 4, increasing the water content of the aqueous mixtures had in all cases the ultimate result of decreasing the efficiency of lignin extraction by the solvent.

The detrimental effect of added water to the organic solvent is probably due to the decreased solubility of the processed biomass in the

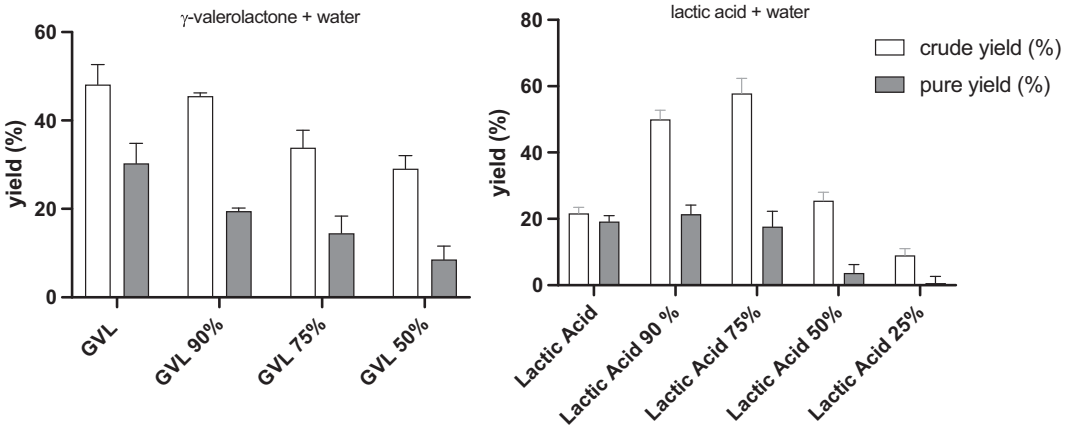


Fig. 4. - Effect of water content (%*w*/w) of two aqueous mixtures (right: lactic acid; left: γ-valerolactone) on the lignin extraction yield. Experimental conditions of extraction:1%w/w sulfuric acid 110 °C, 5 h, 500 rpm.

extracting mixture. Lignin is constituted of chains of polymeric aryl ethers that, upon acidolysis, generate polyphenolic fragments, more soluble in an organic solvent than in water. As a strongly HBD solvent and a nucleophilic species in its own right, water can promote the cleavage of the β-O-4 ethereal linkages in the polymeric biomass. As a solvent, however, it can decrease the solubility of the processed biomass or of its by-products, if compared with the pure organic co-solvent. The extracted lignin comprises a complex mixture of fragments, which may change with the extracting conditions. Pure, simple organic compounds with small molecular weights have solubilities that can be precisely measured in most solvents. Not so with polymeric fragments of an indefinite molecular weight and composition, for which only a speculative suggestion can be made: based on the well-known inmiscibility of aromatic ethers with water, a rather poor solubility of polymeric fragments with the same functionality can be anticipated in water.

The addition of water to the organic solvent has also an effect on changing the microscopic properties of the mixture: its hydrogen-bond-donating (HBD), and –accepting (HBA) strengths, and its dipolarity-polarizability. These solvent properties are frequently associated with changes in the reactivity of a chemical process. Any correlations found between these solvent properties and the efficiency of lignin extraction could therefore suggest the intervention of more specific, microscopic solvent effects into the delignification process.

In order to search for any such correlations, Kamlet-Taft solvent parameters α, β, and π* were estimated for all examined organic solvents and its aqueous solutions, and their values compared with the obtained lignin extraction efficiencies.

3.2. Polarity assessment through solvatochromic measurements

Kamlet-Taft (KT) parameters can be determined for any solvent or solvent mixture. In the latter case, they are obviously the result of the interplay of different factors and components, and the rationalization of the obtained values is more complex than when determined for a single, pure solvent. Nevertheless, they allow a comparison among different mixtures. In the case of aqueous. Mixtures of variable composition, they can reveal specific trends as, for example, how medium properties vary with the addition of a particular component like water. As a tool for the rationalization of solvent effects on chemical reactions, they can be used to interpret the medium effectiveness in the acid-catalyzed cleavage of the ether bonds of the biomass in the pretreatment stage.

Kamlet-Taft solvent parameters for most of the solvents and solvent mixtures were determined with the aid of the solvato- and fluorosolvatochromic dyes of Scheme 2. Their solvatochromic λ_{max} and fluorosolvatochromic λ^{em}_{max} values in all solvent mixtures are listed in Table S1 in the Supporting Information.

Determination of the β and π* values was carried out by the standard comparison of the solvatochromic shifts of 4-nitroaniline (NA) and *N,N*-diethyl-4-nitroaniline (DENA). Parameter α, that measures the medium hydrogen-bond-donating (HBD) strength, is frequently determined with the aid of Reichardt's betaine and the $E_T(30)$ polarity scale, which reflect the acidity α and the dipolarity-polarizability π* of the medium [39–41].

However, in our case, direct measurement of $E_T(30)$ values could not be done, because the strongly acidic mixtures protonate the phenolate probe and eliminate its solvatochromic response. Alternatives to this limitation had been proposed before, by employing other solvatochromic dyes, like Nile Red (NR) [42–44] or PRODAN [45,46] which should be poorly or not affected by an acidic milieu.

Prior to adopting any of these dyes for our analysis, we compared their response in organic solutions before and after addition of 1% sulfuric acid. Nile Red solutions revealed significant bathochromic shifts of the solvatochromic band of the dye, brought about by the addition of acid. Furthermore, a second solvatochromic band also appeared at larger λ_{max} values (Fig. 5). The effect of the acid on the fluorosolvatochromic band of PRODAN was not significant, although in some cases, quenching was observed (Table S3 in the Supporting Information).

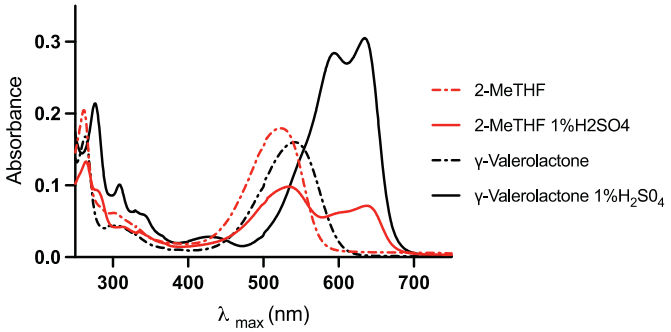


Fig. 5. - Spectra of Nile Red (50 mM) in 2-MeTHF and γ-valerolactone, in the absence and the presence of 1% *w*/w sulfuric acid.

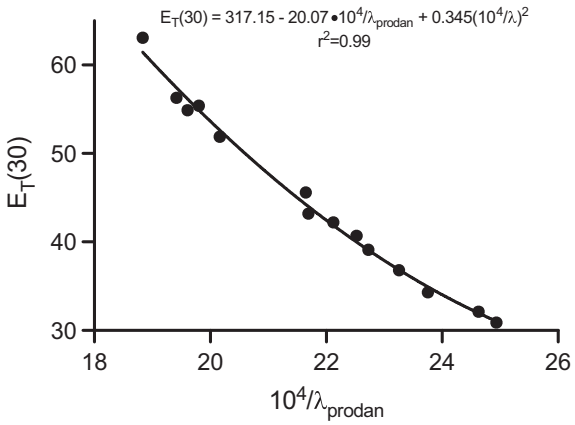


Fig. 6. - Correlation between $E_T(30)$ [47] and ν(PRODAN) [45,46] values of 14 protic and non-protic solvents.

In order to choose the best option between the two solvatochromic probes (Nile Red and PRODAN), we compared correlations of their solvatochromic transition energies with $E_T(30)$ values, with data from literature.

A plot of $E(NR) = 28{,}591/\lambda_{max}$ (NR) against $E_T(30)$ values for 17 protic and non-protic solvents did not exhibit a smooth correlation, but instead yielded two reasonably parallel straight lines for the two sets of solvents [42]. Employing data from the same source and additional $E_T(30)$ values [47] for a total of 14 protic solvents, a straight line with significant dispersion was obtained, with $r^2 = 0.77$ (see Supporting Information, Fig. S7).

By contrast, a plot of $E_T(30)$ values [47] against $\nu\,(PRODAN) = 10^4/\lambda^{em}_{max}$ (PRODAN) for a total of 14 protic and non-protic solvents [45,46] yielded a smooth, almost linear correlation, with a second-order polynomial fitting equation and $r^2 = 0.99$ as shown in Fig. 6.

These results, together with the smaller sensitivity of PRODAN to an acidic milieu, led us to choose this dye to estimate K-T α values of the aqueous mixtures. By employing fitting Eq. (1), we were able to convert $\nu(PRODAN) = 10^4/\lambda^{em}_{max}$ (PRODAN) into equivalent $E_T(30)$ values for all solvent mixtures of Table 1. With these values, and those obtained from the two 4-nitroaniline probes ν(NA) and ν(DENA), K-T parameters α, β and π* were calculated with the aid of Eqs. (2)–(4) [41],

$$E_T(30) = 317.15 - 20.07\nu(PRODAN) + 0.345\,(\nu(PRODAN))^2 \quad (1)$$

$$\beta = (1.035\nu(DENA) - \nu(NA) + 2.64))/2.68 \quad (2)$$

$$\pi^* = (27.52 - \nu(DENA))/3.182 \quad (3)$$

Table 2
– Kamlet-Taft solvent polarity values of aqueous mixtures of variable composition of lactic acid and γ-valerolactone.

Solvent mixture	α_1 [a]	α_2 [b]	β	π*
Lactic acid	–	–	0.29	1.13
Lactic acid + water (10% *w*/w)	0.61	0.63	0.39	1.15
Lactic acid + water (25% w/w)	0.66	0.70	0.36	1.19
Lactic acid + water (50% w/w)	0.66	0.71	0.31	1.26
Lactic acid + water (75% w/w)	0.64	0.70	0.23	1.32
γ-Valerolactone	0.21	0.14	0.46	0.84
γ-Valerolactone + water (10% w/w)	0.60	0.57	0.51	0.96
γ-Valerolactone + water (25% w/w)	0.65	0.64	0.50	1.01
γ-Valerolactone + water (50% w/w)	0.67	0.68	0.48	1.08

[a] Calculated with Eq. (4).
[b] Calculated with Eq. (5).

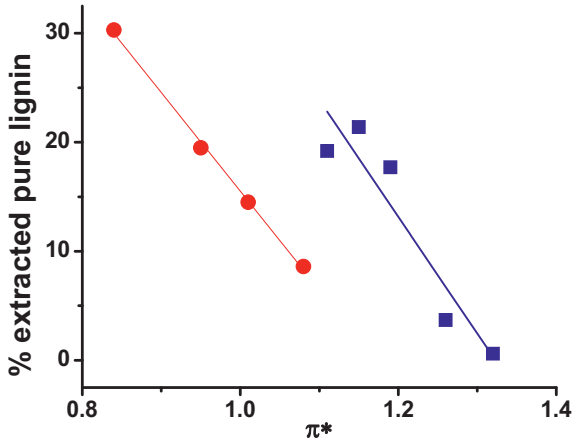


Fig. 7. – Dependence of the lignin extraction efficiency on KAT dipolarity-polarizability π* value of aqueous acidic mixtures of γ-valerolactone (red circles) and lactic acid (blue squares).

$$\alpha_1 = (E_T(30) - 14.6(\pi^* - 0.23) - 30.31)/16.5 \quad (4)$$

Alternatively, Eq. (5) [39] yielded values of α similar to those calculated with Eq. (4).

$$\alpha_2 = 0.0649E_T(30) - 2.03–0.72\pi^* \quad (5)$$

All of these values are listed in Table 2.

Correlations were sought between the percentage extractions of pure lignin by the solvent mixtures (Table 1) and the Kamlet-Taft parameters of Table 2. No correlations were found between parameters α_1 and α_2, or β and the extraction efficiencies (%). By contrast, reasonably linear dependences were observed between the extraction efficiencies and the dipolarity-polarizability parameter π*. In all cases, extraction efficiencies decreased with the increased dipolarity-polarizability of the medium (Fig. 7).

It may seem surprising that no correlation was found with parameter α, which is a measure of the hydrogen-bond-donor strength of the medium, and therefore of its acidity. The initial hypothesis of this work, that aqueous mixtures of acids should promote delignification, would suggest some correlation between α and the extraction efficiency. One should however bear in mind Kamlet and Taft's observation that "hydrogenbonding phenomena involving partial proton transfer (…) have little in common with pK's (proton-transfer measures) when substrates with different functional groups are compared". In fact, α values for ethanol, methanol or water are greater than for acetic acid [48].

The negative effect of the increased medium dipolarity-polarizability on the delignification process can be explained by consideration of the postulated mechanism of acid cleavage of the β-O-4 linkage in lignins. It involves an initial protonation of the ethereal oxygen, followed by a solvent-assisted, nucleophilic displacement of an alkyl fragment. The corresponding transition-state is less polar than the starting protonated ether, and its formation should therefore be rendered more difficult in increasingly polar media (Scheme 3).

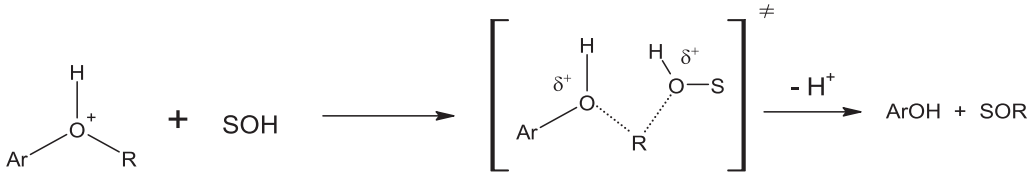
The above analysis must be purely qualitative. The interpretation put forward to explain the observed correlation with the polarizability-dipolarity term of the KT approach must also be regarded with caution. Unlike other works on the preferential solvation of simple solutes like limonin, [49] thiamphenicol [50] or tyrosine [51] employing the KT framework, application of the same approach to a polymeric biomass is prohibitively complex, severely limiting a more quantitative interpretation of the obtained results.

4. Conclusions

The efficiency of lignin extraction from *Pinus radiata* sawdust was evaluated for various organic solvents, acidified with 1% sulfuric acid, and for some of their aqueous mixtures with variable water content. Organic co-solvents included acids and non-acidic products. Solvents could be grouped according to their high (ca. 30% extraction by γ-valerolactone, propylene carbonate and ethylene glycol), regular (20% extraction by lactic acid) and poor performance for wood delignification (<10% extraction by glycerol, 2-methyltetrahydrofuran, acetic and citric acids). IR spectral analysis of the extracted lignin revealed impurities associated to the solvent. Estimation of the purity of the extracted lignin was achieved by measuring the absorbance of the UV–vis spectra of the aromatic polymer at 280 nm. Contamination of isolated lignin varied with the employed solvent and increased with the amount of water added to the organic solvent. A possible source of contamination by solvent by-products could be due to decomposition and polymerization reactions undergone by the solvent under the acidic high-temperature extraction conditions.

The detrimental effect of the added water to the solvent was ascribed to the poorer solubility of lignin in water. Lignin extraction decreased with the increased medium polarity, as expressed by the value of the dipolarity-polarizability parameter π* of the aqueous mixtures. The obtained good correlations between these values pointed to an additional negative effect of the added water on the delignification process. By considering the postulated mechanism of acid-catalyzed cleavage of the ethereal β-O-4 linkage in lignin, a transition state less polar than the protonated aryl ether would not be favored in the increasingly aqueous polar environment.

The present work highlighted the negative effect of water on the delignification process by aqueous solvent mixtures, in spite of its expected contribution to the hydrolysis and fragmentation of the



Scheme 3. - Schematic mechanism for the acid cleavage of the β-O-4 linkage in lignins, comparing the relative polarity of the reacting protonated ether with that of the transition state.

polymeric biomass. It also shed light on the problems of lignin contamination by solvent degradation, not always addressed in reports of lignin extraction. The use of KT parameters for the assessment of the polar properties of the employed solvents led to an interesting correlation with the dipolarity-polarizability of the medium. This piece of information, together with the reported efficiencies of solvents like γ-valerolactone, propylene carbonate and ethylene glycol, can be used for the design of new solvents. Our choice of several organic solvents in designing aqueous mixtures was inevitably limited. More systematic work is needed in this line with a larger set of solvents, of related or similar functionalities to those tested in this work, capable of a more efficient lignin extraction with less product contamination.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2022.05.030.

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